

Evidence review

Twenty-seven measured cells across two method arms and five datasets, read against what each one can actually establish. Several conclusions carried through the week do not survive the full table; they are corrected here rather than quietly dropped.

arms: RePaint-part · joint-6D datasets: ours · 3DCoMPaT200 · DensePoint · PartNeXt

source: W&B `colored-pc-completion`

1 · The ledger

Across all 27 cells of the RePaint arm, counting how often each method beats the **NN-copy** baseline — copy the colour of the nearest visible point:

REPAINT-VANILLA (PART-BLIND) 0 / 27 Never wins. Diffusion alone, without part information, loses in every single cell.	PART-MEAN (ORACLE LABELS) 12 / 27 A flat per-part average beats the baseline in nearly half the cells — the part label carries real information.
REPAINT-PART (ORACLE LABELS) 5 / 27 All five sit in the two configurations where the part-constant ceiling is far below NN-copy. That is not a coincidence — see §5.	REPAINT-PART (TRAINED SEGMENTER) 3 / 27 Two clear wins, both <code>ours/airplane/long4k</code>. The third is a 0.10 dE margin and should not be counted.

THE ONE CLEAN RESULT

Part-blind diffusion never beats the trivial baseline — 0 out of 27. Adding part conditioning moves the same architecture to 5 wins with accurate labels. Whatever else is uncertain, the gain in this system comes from part awareness, not from diffusion.

2 · Most rows are not evidence

The runs were not made under one configuration. Reading them as a single table is the mistake that produced two wrong conclusions this week.

EXP TAG	DDPM EPOCHS	COMP_POINTS	FRAME VERDICT	STATUS
base	300	none	B0ZUK	Not usable. Three defects at once: short training, the density-shift bug unfixed, and PoinTr failing to recognise the shape at all
long4k	4000	2048	OK	Valid. The only sound <code>ours</code> configuration — and it covers airplane only
multi3	300	2048	OK / ZAYIF	Undertrained. Geometry fixed, but at the training budget already shown to be insufficient

Every `ours/car` and `ours/chair` row is `base`. Their numbers — chair at 46.43 dE against a baseline of 12.92 — describe a broken configuration, not a method.

3 · Training budget flips the sign

Same category, same data, same code. Only the configuration differs:

OURS / AIRPLANE	EXP	EPOCHS	NN	REPAINT PART	MARGIN
simple	base	300	18.15	22.25	+4.10
simple	long4k	4000	19.19	14.90	-4.30
moderate	base	300	21.63	25.38	+3.74
moderate	long4k	4000	23.93	19.95	-3.98
hard	base	300	18.23	27.84	+9.61
hard	long4k	4000	19.71	22.70	+2.99

A 7.3 dE swing at the two lower difficulties, enough to reverse the conclusion. The **+9.61** figure carried through most of this week is the `base/hard` cell — a number from the unusable configuration.

WHAT THIS MEANS FOR THE THREE NEW DATASETS

The `multi3` sweep — 3DCoMPaT200, DensePoint and PartNeXt — ran at **300 epochs**. That is the budget just shown to invert the result on data where the method does work. Those rows cannot be read as a verdict on part conditioning. They can still be read for what §5 and §6 extract from them, because those arguments run through the oracle rows, which do not depend on the trained denoiser.

4 · All 27 cells

Mean ΔE on the missing region; lower is better. **teal** marks a method beating NN-copy. Greyed rows are the unusable **base** configuration.

DATASET / CATEGORY	DIFF	NN-COPY	PART MEAN	VANILLA	REPAINT-PART	ORACLE SEG	CEILING	SEG
3dcompat / chair multi3	simple	19.94	25.38	27.59	21.88	13.37	5.07	57.1%
	moderate	21.63	28.55	31.65	23.46	13.66	5.30	57.1%
	hard	28.45	32.24	34.57	28.35	21.57	5.45	57.1%
densepoint / chair multi3	simple	19.64	20.32	30.27	24.80	24.38	20.01	85.5%
	moderate	23.05	21.68	33.68	27.62	27.91	20.16	85.5%
	hard	23.34	23.23	36.64	34.69	31.89	19.99	85.5%
ours / airplane	simple	18.15	15.43	28.45	22.25	22.17	14.17	85.5%
	moderate	21.63	19.79	30.48	25.38	25.67	16.74	85.5%
	hard	18.23	19.95	33.48	27.84	28.29	17.29	85.5%
ours / airplane long4k	simple	19.19	17.66	20.31	14.90	15.16	15.37	85.0%
	moderate	23.93	21.96	26.93	19.95	19.58	18.38	85.0%
	hard	19.71	21.24	27.63	22.70	21.26	18.32	85.0%
ours / car	simple	29.62	34.59	48.74	40.53	39.94	31.72	80.5%
	moderate	29.73	34.90	51.17	44.26	44.08	31.08	80.5%
	hard	34.88	35.54	51.15	48.82	47.58	31.39	80.5%
ours / chair	simple	11.82	14.09	59.53	35.54	38.21	11.74	84.7%
	moderate	11.67	13.81	64.97	44.67	45.73	11.47	84.7%
	hard	12.92	14.75	80.83	46.43	49.93	11.34	84.7%
ours (early) / airplane	simple	20.33	18.56	37.90	29.51	30.13	16.65	85.0%
	moderate	22.62	20.60	41.30	32.92	33.23	17.47	85.0%
	hard	23.03	22.03	42.03	36.84	36.99	17.54	85.0%
ours (early) / car	simple	31.12	34.96	50.53	57.71	58.11	32.61	83.2%
	moderate	31.99	35.34	47.99	62.22	62.21	32.37	83.2%
	hard	36.96	35.42	48.44	62.14	62.19	32.60	83.2%
partnext / chair multi3	simple	7.77	10.52	21.15	17.20	18.86	8.24	73.8%
	moderate	11.07	9.58	33.52	20.73	21.41	8.33	73.8%
	hard	9.92	9.51	39.51	24.95	27.17	7.58	73.8%

5 · The ceiling predicts what is possible

The **oracle ceiling** paints every point its own part's true mean colour. No part-constant method can score below it. Comparing that ceiling to NN-copy says, before any model is trained, whether part information can help at all in a given cell.

margin = oracle ceiling - NN-copy (negative = parts carry usable signal)

MARGIN	CELLS	ORACLE-SEG BEATS NN	MEAN ORACLE-SEG RESULT
ceiling more than 8 dE below NN	3	3 / 3	-7.14 dE
between 2 and 8 dE below	13	2 / 13	+8.69 dE
less than 2 dE below, or above	11	0 / 11	+18.80 dE

Monotone, and the correlation across all 27 cells is $r = 0.61$. The rule is one-directional and that is how it should be used: **a large margin does not promise a win, but a small one rules it out**. Where the ceiling cannot beat NN-copy, nothing built on part-constant colour can.

DATASET / CATEGORY	CEILING	NN-COPY	MARGIN	CONSEQUENCE
3DCoMPaT200 / chair	5.07	19.94	-14.87	Oracle labels beat NN by 6.6 dE — the largest margin measured anywhere
ours / airplane long4k	15.37	19.19	-3.82	The only cells where the trained pipeline wins
ours / chair	11.74	11.82	-0.08	No usable signal — NN-copy is already at the ceiling
DensePoint / chair	20.01	19.64	+0.37	Ceiling is worse than the baseline. More training cannot fix this
PartNeXt / Chair	8.24	7.77	+0.47	Same — artist textures cut across the derived part vocabulary

6 · Segmentation cost tracks vocabulary size

The gap between RePaint-part and RePaint-part (oracle seg) is what the trained segmenter costs. It is negligible everywhere except one place:

DATASET / CATEGORY	SEG ACC	PART VOCABULARY	COST	READING
3DCoMPaT200 / chair	57.1 %	22	+8.37 dE	The bottleneck. Oracle labels win by 6.6; the segmenter turns that into a 1.9 loss
PartNeXt / Chair	73.8 %	7	-1.52 dE	Trained labels marginally better than oracle — within noise
ours / airplane long4k	85.0 %	4	+0.51 dE	Segmentation is not the limit here
DensePoint / chair	85.5 %	4	+0.98 dE	Also not the limit — the ceiling is

This separates two failure modes that look identical in the headline number. On DensePoint the method fails because *the data has no part-colour signal*. On 3DCoMPaT200 the signal is the strongest we have measured and the method fails because *it cannot find the parts*. Only the second is fixable by better modelling.

7 · The joint-6D arm

Geometry and colour from one diffusion model, no frozen PoinTr. Nine cells, complete.

- **Colour loses in 9 / 9**, by $+4.06 \pm 0.94$ dE against NN-copy evaluated on the same generated geometry. Low variance across category and difficulty — systematic, not noise.
- **Geometry is competitive.** Chamfer 0.044 at simple against frozen PoinTr's 0.054; PoinTr pulls ahead at hard (0.073 vs 0.089).

- **Completion only pays off under heavy occlusion.** The ratio to returning the input unchanged is 0.98–1.15 at simple, rising to 2.4–3.9 at hard. Both arms show this.

VERDICT WITHHELD

The ablation separating the single-sample penalty and the correspondence confound has not been run. Until it is, +4.06 cannot be attributed to the model rather than to the measurement — ΔE rewards the conditional mean, and a sampler pays variance for it. The notebook is ready and takes about an hour.

8 • Corrections

CARRIED THIS WEEK	CORRECTED
“RePaint-part loses by $+9.61 \pm 3.04$ ”	That is the <code>base/hard</code> cell — unusable configuration. The valid figure is +2.99, and at simple and moderate the method <i>wins</i>
“Part conditioning wins”	Over-claimed from three airplane rows. It wins in 2 of 27 cells with a trained segmenter, 5 with oracle labels. What holds without qualification is the negative: part-blind diffusion wins 0 of 27
“The fixes flipped the result”	The training budget did most of the work: 300 → 4000 epochs, alongside the frame and density fixes
DensePoint has “no headroom”, so drop it	Right conclusion, and now confirmed on the real run — its ceiling is 0.37 dE worse than NN-copy. But it was dismissed too early on a comparison with our own car, which is equally thin
<code>headroom.py</code> measures usable signal	It compares the oracle to a <i>global mean</i> , a much weaker baseline than NN-copy, so it overstates. Our chair reads 3.85 there but its real margin is −0.08. The script needs an NN-copy comparator

9 • What is established, and what is not

Established

- Part-blind diffusion does not beat nearest-neighbour copying — 0 of 27 cells, across five datasets.
- With accurate part labels and sufficient training, part conditioning beats it, and can exceed the part-constant ceiling: 14.90 against a ceiling of 15.37 on `ours/airplane/simple`.
- The ceiling-to-NN margin, measurable before training, rules out cells where no part method can win.
- DensePoint and PartNeXt chair carry no usable part-colour signal. This is a property of those datasets and is reportable as such.
- Segmentation is the binding constraint exactly once — 3DCoMPaT200, 22 parts, 57.1 % accuracy, 8.37 dE cost.

Not established

- Whether the win generalises beyond airplane. Car and chair have never been run in a valid configuration.
- Whether the joint-6D loss is real or a measurement artefact. The ablation is unrun.
- Whether 3DCoMPaT200 confirms the claim. Its oracle rows are promising, but the trained result is at 300 epochs.
- Anything about statistical significance. Cells are means over 15 test models, one seed, no test.

10 • The three runs that would settle it

#	RUN	COST	WHAT IT DECIDES
1	<code>ours/car</code> + <code>ours/chair</code> at long4k	~6 h	Whether the airplane win is the method or the category. Currently the single largest gap

#	RUN	COST	WHAT IT DECIDES
2	joint-6D ablation	~1 h	Whether +4.06 is the model or the metric. Cheapest decisive run available
3	3dcompat at 4000 epochs	~3 h	Whether the 6.6 dE oracle margin survives a trained segmenter. Not worth running for DensePoint or PartNeXt — their ceilings already lose

docs/results.html · pulled from W&B project colored-pc-completion, 43 runs, 27 evaluated cells · regenerate the PDF with Chrome headless --print-to-pdf